

# INTERN FINAL PRESENTATION

*A DevOps Summer Recap*



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# FROM NOTEBOOK TO PIPELINE

*Twelve weeks of a DevOps intern summer from start to finish.*

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# Where I Started

*Same instinct for operational visibility, aimed at a very different problem this time.*



## Primary Goal

- Last summer: migrate AWS CloudWatch dashboards to Grafana
- This summer: build a CI/CD workflow to deploy AWS resources using custom actions, base templates, and reusable workflows



## Core Structure

- YAML both summers
- Last summer: Azure DevOps (AzDO)
- This summer: GitHub Actions
- Also learning AWS Cloud from the ground up



## Security & Reliability

- OIDC and role ARNs in place of hardcoded keys
- Researched DevOps scanning tools to better understand their capabilities.
- Received mentorship on SAST scanning, Dependabot, and dependency management best practices.

Workflow anatomy: workflows → jobs → steps → runners, base templates

# Building the Foundation

Weeks 4-6



## CloudFormation Templates

Parameters, Resources, Mappings, and Outputs — modeling dependencies and managing drift across environments.



## OIDC Trust Relationships

GitHub as identity provider, a role trust policy scoped to org/repo, and short-lived token exchange instead of long-lived secrets.



## Security & Dependency Scanning

Gained exposure to SAST scanning using DevOps Tool Scan (CodeQL), along with Dependabot and dependency management best practices.



## GitHub Actions Architecture

Event → Workflow → Job → Runner → Steps → Actions  
Reusable Workflows + Custom Actions

# AWS Cloud Practitioner, In Practice

*Same four domains — studied for the exam, then put to work on the platform itself.*



## Cloud Concepts

Chose between EC2, Elastic Beanstalk, App Runner, and Lambda based on how much infrastructure control a given workload actually needed.



## Security and Compliance

Put the shared responsibility model into practice — OIDC trust relationships and least-privilege IAM on my side, AWS's managed patching on theirs.



## Cloud Technology and Services

Worked hands-on with S3, Lambda Functions, and Secrets Manager



## Billing, Pricing, and Support

Tracked spend with Cost Explorer and Budgets, and compared Savings Plans against Reserved Instances for the platform's support tier.

# Mid-Summer & the Hackathon

*Weeks 7 and 8 — the mid-summer presentation and the Caterpillar Global Hackathon.*





# Delivering the Mid-Summer Story

*Manager feedback shaped how — and whether — the technical work got told.*

- CloudFormation templates tested and validated end-to-end ahead of the demo
- 1:1 with Vinit on how to structure the story and what to highlight
- Delivered “Beyond the Script” — the case for agentic DevOps at Caterpillar
- AWS practice exam scores climbing into cert-ready range
- Contributed to the development of both a web and mobile application for a hackathon
- Shifted my mindset from solely learning to balancing both learning and development

## THE FEEDBACK

*"Grow vertically in your area of ownership while expanding horizontally across the DevOps team to build a well-rounded understanding of the organization."*

*The presentation itself became a deliverable — not just the pipeline behind it.*



# Team 9 — 20/20 VisionLink

*A recommendation-driven dashboard prototype, built in a single week.*



## My Role

- Product manager — ideated the project from scratch
- Presented the business case to get the team on board
- Built the bulk of the app's functionality — the recommendation system itself



## The Team & Stack

- Eric: frontend · Anokhi: ML backend
- Web: React + TypeScript
- Mobile: native, built in Xcode
- Backend: Ollama
- UI in Figma Make + AI orchestration



## The Build

- Recommendation-based dashboard tracking user interactions
- Secured by two-factor authentication (2FA)
- Alerts fleet managers, Google-style, if someone else views their assets



# CI/CD Platform & the Pivot

*Weeks 9 and 10 — the first working demo, and the pivot into research of my own.*



# First Demo → First Real IAM

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*Deploying three resources end-to-end — and finally understanding the permissions underneath.*



## The First Demo

- Finished and presented the CI/CD pipeline's first real demo
- An end-to-end deployment of three AWS resources
- Utilized base templates, custom actions, and reusable workflows



## Reverse-Engineering IAM

- Learned to read and reverse-engineer IAM roles directly
- Understanding policies and permissions instead of just following a template
- Finding the difference between CFN and OIDC roles

***The moment the pipeline stopped being copied from a template and started being understood.***

# From Mentorship to My Own Research

*Where the seed of a summer's mentorship started becoming something of my own.*

- Weeks 1–9:
  - Built understanding of GitHub Actions, CloudFormation, OIDC, IAM
  - Carrying lessons forward from last summer's AIOps work
- Week 10:
  - Began implementing Genesis - the parallel with my original LLMOps pipeline research
  - Layered directly onto that same CI/CD workflow
- Not a new project: my initiative to enhance the existing workflow
  - I wanted to find a way to positively influence the team through a future-focused initiative that could provide long-term value beyond my internship.
  - This research ultimately became the foundation for my Emerging Technology Initiative.

***The seed of a summer of mentorship, starting to bloom into research of my own.***



# What's Next

*The people behind the summer, the ask for what comes after, and a look back at the whole journey.*



# Emerging Tech: My Influence

Sherlock

InfraGen

Wiz

## Genesis

- LLMOps workflow that evaluates deployment changes, assesses risk, and routes actions through PROCEED, AUTO-FIX, or REVIEW-PR paths.
- Combines proactive scanning, governed remediation, observability, and AI-driven recommendations to improve deployment safety.
- Enhances Security, CloudOps, and FinOps through faster feedback, automated reviews, and human-governed decision support.

# Built Together Not Independently

*The mentorship that shaped the technical decisions all summer.*



## Deeksha - Mentor

- Explained LTA, Dependabot, Dependency Management and SAST scanning
- Covered custom actions, reusable workflows, CI/CD, and base templates



## Vinit - Teammate

- Deployment and repo structure
- The deep dives that shaped CloudFormation and release design



## Nathan - Manager

- Gave direction on learning vertically
- At each stop, branched out horizontally into CloudOps, FinOps, and Security



## Cora - Hack Mentor

- Hackathon mentor — guided Team 9 to a second-round appearance
- Walked me through her day-to-day in FinOps

Also Sonia, Angie, and Rob — OIDC + HR Advice

# Where I Want to Take This

“My goal in this internship is to understand where the team is today, identify where I can add real value, and execute on that with forward-looking approaches the team has not yet used, so I can contribute not just as an intern, but as someone helping push the work toward what comes next.”

|                   |                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------|
| <b>Vision</b>     | A path that builds on this summer's CI/CD work and the Genesis research, not a restart. |
| <b>Impact</b>     | Fewer manual pipeline hours, safer deployments, a stronger security posture.            |
| <b>Continuity</b> | Day-to-day platform work feeding directly into the bigger picture.                      |





# From Notebook to Deployment



**Week 1**

Onboarding



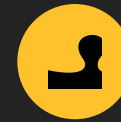
**Week 4**

Foundation



**Week 7**

Mid-Summer



**Week 8**

Hackathon



**Week 10**

Genesis Pivot

*Twelve weeks of GitHub Actions, CloudFormation, and disaster-recovery thinking — turned into a working, documented CI/CD platform.*

Trustan · Questions welcome